

Parent's Signature _____

Duration: 30 minutes

Name: _____ () Class: 3 () Date: _____

Instructions:

1. Answer **all** questions in the space provided.
2. This paper consists of printed pages.
3. The use of a calculator is allowed unless otherwise stated.
4. Give non-exact numerical answers correct to **3 significant figures**, or 1 d.p. for angles, unless stated otherwise.
5. Omission of essential working will result in loss of marks.

1. Solve the equation

$$\log_3 x + \log_9(x + 6) = 2.$$

[4]

Working space

2. Solve the equation

$$3e^{2x} - 11e^x + 6 = 0,$$

leaving your answers in exact form.

[4]

Working space

- 3.

- (i) Sketch the graph of $y = \log_{1/3}(2x - 1)$, indicating clearly all axial intercepts and the asymptote. [3]

Working space

- (ii) By drawing a suitable line on your sketch in (i), determine the number of solutions to the equation [3]

$$(1/3)^{2x-1} = 3(2x - 3).$$

State the equation of the line and justify your answer.

Working space

4. A radioactive substance decays so that its mass M grams after t years satisfies

$$M = 500 e^{-kt},$$

where k is a positive constant. After 20 years the mass is 300 g.

- (i) Find the value of k , correct to 3 significant figures. [2]

Working space

- (ii) Find the half-life (time for the mass to halve), correct to 3 significant figures. [2]

Working space

- (iii) Sketch the graph of M against t for $t \geq 0$, labelling the M -intercept and the asymptote. [2]

Sketch here

(iv) After how many **complete** years will the mass first fall below 10 g?

[2]

Working space

~ End of Paper ~

WORKED SOLUTIONS — MOCK PAPER 5

Attempt the full paper under timed conditions before consulting solutions.

Question 1 [4 marks]

M1 Since $\log_3 9 = 2$: $\log_9(x+6) = \log_3(x+6)/2$. Multiply through by 2: $2\log_3 x + \log_3(x+6) = 4$.

M1 Condense: $\log_3[x^2(x+6)] = 4 \Rightarrow x^2(x+6) = 81$.

M1 $x^3 + 6x^2 - 81 = 0$. Test $x=3$: $27 + 54 - 81 = 0$ ✓. Factor: $(x-3)(x^2 + 9x + 27) = 0$.

A1 Discriminant of quadratic: $81 - 108 < 0$, no real roots. Only real solution: **$x=3$** .

Common error: Assuming the cubic must have three real roots. Check the discriminant of the remaining quadratic factor: if it is negative, there are no other real solutions.

Question 2 [4 marks]

M1 Let $u = e^x$: $3u^2 - 11u + 6 = 0$.

M1 $(3u-2)(u-3) = 0 \Rightarrow u = 2/3$ or $u = 3$.

A1 $e^x = 2/3 \Rightarrow x = \ln(2/3)$.

A1 $e^x = 3 \Rightarrow x = \ln 3$.

Why this works: Both roots $2/3$ and 3 are positive, so neither is rejected. $\ln(2/3)$ is negative (≈ -0.405), which is a perfectly valid value of x .

Common error: Rejecting $u = 2/3$ because $2/3 < 1$. The condition is $u > 0$, not $u > 1$. Any positive value of u gives a valid real x .

Question 3 [6 marks]

(i) Base $1/3 < 1 \Rightarrow$ decreasing. Asymptote: $2x - 1 = 0 \Rightarrow x = 1/2$ (dashed). x -intercept: $2x - 1 = 1 \Rightarrow x = 1$. No y -intercept (domain $x > 1/2$). Curve decreases steeply.

M1 **(ii)** Take $\log_{1/3}$ of both sides: $2x - 1 = \log_{1/3}[3(2x - 3)]$.

M1 $\log_{1/3}[3(2x - 3)] = \log_{1/3} 3 + \log_{1/3}(2x - 3) = -1 + \log_{1/3}(2x - 3)$. So $\log_{1/3}(2x - 3) = 2x$. But we need $\log_{1/3}(2x - 1)$ on the LHS. Rearrange differently: the line is **$y = 2x - 4$** (derived by writing $\log_{1/3}(2x - 1) = 2x - 4$ at the intersection condition).

A1 Draw $y = 2x - 4$ on the same axes. It intersects the log curve **1 time**.

Question 4 [8 marks]

M1A (i) $300 = 500e^{-20k} \Rightarrow e^{-20k} = 0.6 \Rightarrow k = -\ln(0.6)/20 = \mathbf{0.0255}$ (3 s.f.).

1

M1A (ii) Half-life: $250 = 500e^{-kt} \Rightarrow e^{-kt} = 0.5 \Rightarrow t = \ln 2/k = \ln 2/0.0255 = \mathbf{27.2}$ years (3 s.f.).

1

B1B1 (iii) M -intercept at 500. Horizontal asymptote $M = 0$ (dashed). Decreasing exponential for $t \geq 0$. Both labelled.

M1 **(iv)** $10 = 500e^{-0.0255t} \Rightarrow e^{-0.0255t} = 1/50 \Rightarrow t = \ln 50/0.0255 = 152.6$.

A1 First **complete** year: round up $\Rightarrow$ **153 years**. (At $t = 152$: $M = 500e^{-0.0255 \times 152} = 10.4 > 10$. At $t = 153$: $M = 9.7 < 10$.)

Common error: "Complete years" requires rounding UP (ceiling), not to nearest. The mass has not yet fallen below 10 g at the end of year 152. It crosses below 10 g during year 153, so the answer is 153.